

TECHNOLOGY CONSULTING

IN THE GLOBAL COMMUNITY

Final Consulting Report

Palau FIU

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Carnegie Mellon University



The Republic of Palau Financial Intelligence Executive Summary

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I. About the Organization

The Republic of Palau Financial Intelligence Unit (FIU) is an independent agency established in 2001 under the Money Laundering and Proceeds of Crime Act, made fully autonomous by a 2014 amendment. Headquartered in Koror, its mandate is to detect, disrupt, and deter money laundering, terrorism financing, and related transnational crime by analyzing and disseminating financial intelligence, and supervising Financial Institutions (FIs) and Designated Non-Financial Businesses and Professions (DNFBPs) for AML/CFT compliance.

The mission statement of the FIU is,

Detect, disrupt, and deter money laundering, terrorism and proliferation financing, and other transnational crimes through coordination, collaboration, and cooperation with relevant domestic and international partners, while working towards alignment with international best practices and standards and promoting FIU staff growth and excellence.

The vision statement of the FIU is,

To safeguard the Republic of Palau against illicit financial activity.

II. Digitization of A Bank's Legacy CTRs

Prior to 2015, banks submitted CTRs exclusively on paper. While past consultants established the process of digitizing other banks' records, one bank's complex format remained unprocessed.

Outcomes

- Enabled digitization of complex paper-based CTRs through a newly developed pipeline
- Successfully processed real CTRs with FIU staff and transferred technical knowledge to Boboy
- Validated end-to-end operation of the pipeline and documented system setup for future sustainability

Outputs

- A Python-based system that extracts structured data from scanned PDF files
- Provided detailed documentation and codebase for independent use and future updates

III. AI-Powered Web Application for Compliance Report Review

FIU conducts periodic on-site examinations of FIs and DNFBPs based on risk ratings. Reviewing compliance reports manually is time-consuming and prone to human error. An AI-powered solution was developed to streamline this process.

Outcomes

- Improved the efficiency and consistency of compliance reviews

- Tested the system with the intelligence analyst and compliance officers

Outputs

- Delivered a FastAPI based web system with LangChain integration
- Provided training walkthroughs, source code, and documentation for continued internal use

IV. Digital Forms for Compliance Information Submission

The FIU previously relied on Excel spreadsheets to collect compliance data from FIs and DNFBPs, leading to inefficiencies in data management.

Outcomes

- Reduce manual processing and ensured structured data collection from reporting institutions
- Enabled FIU staff to independently create and manage new forms going forward

Outputs

- Created four digital forms for FIs and DNFBPs, two of which are currently on Zoho Forms
- Delivered training videos to ensure sustainable management by FIU staff

V. New Database Data Migration and Setup

FIU adopted a new database to store CTRs and STRs in 2024 for managing financial intelligence. Support was required to migrate historical data and configure the system for ongoing use.

Outcomes

- Resolved key setup and configuration issues that blocked data imports
- Reintroduced the system into daily FIU operations

Outputs

- Fully configured the new database system with migrated historical data

VI. Recommendations

Upgrading from a CPU-only server to a GPU-enabled server would significantly enhance the performance and usability of AI tools developed this summer.

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Yang is a junior in Information Systems.
 He will come back to Carnegie Mellon as a senior student next fall.

The Republic of Palau Financial Intelligence Unit
<https://www.palaufiu.org>

About the Consultant

Yang Pan

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I. About the Organization

Organization

The Republic of Palau Financial Intelligence Unit (FIU) is an independent government agency established in 2001 under the Money Laundering and Proceeds of Crime Act and made fully autonomous by the 2014 amendment. Headquartered in Koror, Palau, the FIU’s statutory mandate is to detect, disrupt, and deter money-laundering, terrorism-financing, and related transnational crime by receiving, analyzing, and disseminating financial intelligence, and by supervising both Financial Institutions (FIs) and Designated Non-Financial Business and Professions (DNFBPs) for Anti-Money Laundering and Counter-Terrorist Financing (AML/CFT) compliance.

The mission statement of the FIU is,

Detect, disrupt, and deter money laundering, terrorism and proliferation financing, and other transnational crimes through coordination, collaboration, and cooperation with relevant domestic and international partners, while working towards alignment with international best practices and standards and promoting FIU staff growth and excellence.

The vision statement of the FIU is,

To safeguard the Republic of Palau against illicit financial activity.

To fulfil this mandate, the FIU gathers data from five retail banks: Bank of Hawaii, Bank of Guam, Bank Pacific, Asia Pacific Commercial Bank, and Palau Investment Bank, as well as the Bureau of Customs and Border Protection. These entities file Currency Transaction Reports (CTRs) for any cash transaction above \$10,000 and Suspicious Transaction Reports (STRs) whenever activity appears abnormal or potentially illicit.

The organization operates with a current budget of \$227,000. Of this, about \$160,000 is allocated to salaries and rental expenses, \$10,000 to the Asia/Pacific Group (APG), and the remainder is used for operations such as equipment and business travel. Director Fincher manages these funds while serving a wide range of partner agencies, from the Office of the Attorney General to the Digital Residency Office.

Facilities

The FIU shares office space with the Financial Institutions Commission (FIC) on the second floor of the Surangel and Sons Co. building in Koror. These two entities collaborate closely to oversee financial activities within Palau. The office features six office rooms, one meeting room, and a shared kitchen. Power and climate controls are adequate, and doors are secured after hours to protect both FIU and FIC workspaces.

Programs

A key operational focus for the FIU is converting paper-based transaction reports into digital formats, facilitating rapid analysis and improved investigative efficiency. Prior to 2015, data

management was challenging due to reliance on physical documents. The digitization effort, initiated in 2015, now allows direct electronic submission from financial institutions, significantly enhancing the FIU's analytic capabilities.

Data received electronically in CSV format undergoes standardized processing by Senior Analysis Boboy, who uploads and organizes these files into the FIU's Microsoft Access database. The recently installed database system complements this setup, offering advanced analytics and visualization capabilities through graph-based analysis.

Staff

Adair K. Fincher serves as Director of the FIU. She sets strategy, oversees the budget, manages staff and contracts, and represents Palau at regional and international AML/CFT forums. Day-to-day, she relies on Zoho for email, and Microsoft Office for document drafting and analysis; she does have log-in credentials for the FIU's Microsoft Access database but uses it only for occasional look-ups rather than routine data work.

Hussein "Boboy" Derbai, the Senior Intelligence Analyst, is the FIU's principal technologist. He cleans, standardises, and uploads CTR and STR files to the Access database, designs ad-hoc queries, and prepares intelligence briefs for the Director. Because of his hardware background, Boboy also handles most first-line troubleshooting for the office's PCs and network gear.

Jeremy Lee is the AML/CFT Officer for Designated Non-Financial Businesses and Professions (DNFBPs) and Virtual Asset Service Providers (VASPs). He plans and conducts onsite and off-site examinations, drafts compliance reports, and recommends enforcement actions when necessary. Jeremy is currently shadowing Boboy to learn the Access workflows so he can run basic queries and assist with data uploads.

Kara K. Remeliik is the Compliance Officer responsible for licensed Financial Institutions (FIs). She schedules and performs supervisory exams, writes findings, develops training materials, and coordinates with the Financial Institutions Commission on enforcement.

Adair, Boboy, and Jeremy have user accounts on the Microsoft Access database; however, Boboy remains the primary operator and is actively training Jeremy to share that workload.

Technology Infrastructure

The office has four Windows PCs and a server on Ubuntu Mate, which hosts the AI service and the new database system. A Synology NAS provided local file storage and sharing. A network printer/scanner is shared across the LAN.

Internet arrives via the under-sea cable, delivering roughly 9 Mbps down and 11 Mbps up, sufficient for large-file transfers and remote meetings.

Technology Management

The FIU operates with a small team of four members, so technology management remains largely ad hoc. Boboy handles day-to-day fixes, and more complex issues go to Cjay Shiro's private tech-services firm, who previously served as FIC's executive commissioner. When necessary, Bel-Pacific supplies an on-call specialist. Backups occur to external drives because cloud services remain outside the current budget.

Technology Planning

Boboy drafts IT needs, and the final decisions are made by Adair. Roughly 5% of the overall budget is earmarked for equipment freshes. Both leaders stay current on AML/CTF tech by attending international conferences and comparing notes with peer FIUs.

Communication

Internal files move through email, the Synology NAS, and face-to-face discussions. External correspondence relies on password-protected email and phone. Bandwidth currently supports secure transfer of large CSVs and PDF intelligence packages.

Information Management

The primary Access database, originally built with TCinGC support, stores CTR and STR data supplemented by World-Check and Sayari for background checks. A new database system, added 2024, offers network-analysis views, and two rotating off-site hard-drive backups protect the data. The FIU plans to add a secure portal for form exchange with other law-enforcement bodies, and a case-management database for intelligence investigations, but it requires collaboration from all other agencies, and no progress has been made.

Business Systems

Hiring begins with the Ministry of Human Resources, which forwards resumes to the FIU for interviews and selection. Payroll is processed centrally through the Bureau of National Treasury under the Ministry of Finance.

II. Digitization of A Bank's Legacy CTRs

Motivation

Banks in Palau are required to submit Currency Transaction Reports (CTRs) to FIU by law. Prior to 2015, CTRs were submitted exclusively in paper form. While previous TCinGC consultants created processes to digitize CTRs from two banks, the reports from another bank remained untouched due to complex formatting and layout challenges.

This limitation impacts the FIU's mission significantly. Without searchable access to historical financial transactions, compliance officers are unable to easily trace long-term cash flow trends, conduct longitudinal analyses, or contextualize new Suspicious Transaction Reports (STRs). Digitization is critical for enhancing investigative capabilities and ensuring that historical data continues to support ongoing AML/CFT enforcement.

Outcomes

A detailed analysis of this bank's CTRs was conducted to identify the specific layout challenges, such as text boxes and underlines, that caused previous OCR tools to fail. I also reviewed the scripts developed by previous consultants to understand their limitations.

A new Python-based digitization pipeline was developed to process the complex CTR forms from this bank. The pipeline intelligently combines two key components: (1) traditional OCR (using Tesseract) to extract basic layout features such as page numbers, and (2) a vision-based large language model (LLM) hosted locally via Ollama, capable of interpreting complex layouts, such as handwritten text and underlined labels, to accurately extract structured data.

The pipeline processes documents in stages:

1. Preprocessing scanned PDFs into image files
2. Running OCR to extract page numbers to separate them into individual documents
3. Segmenting document regions for efficient model input
4. Feeding image regions into the LLM, with prompts tailored to the form's visual structure
5. Formatting the extracted output into structured CSV files for analysis

To ensure sustainability, the source code and detailed documentation on system setup, file organization, and how to upgrade LLM are delivered to FIU.

The project provided the FIU with a functioning digitization pipeline for the previously inaccessible CTRs, converting them from paper forms into structured, searchable data. During the engagement, I worked closely with Boboy to test and validate the tool on real scanned CTR samples. Boboy has successfully processed one document end-to-end, demonstrating the pipeline's effectiveness, and he was also trained on how to run the system independently, including how to organize files and execute the script. The FIU now has the technical knowledge and documentation to operate the pipeline without further assistance.

While the solution is functionally successful and has been tested end-to-end by FIU staff, its long-term adoption may be limited by a significant performance bottleneck. The pipeline relies on an LLM running on the FIU's current CPU-only server. This hardware is not optimized for AI workloads, resulting in slow processing times. Processing a single scanned CTR PDF currently takes about 10 minutes. This slow speed could discourage regular use and create a backlog if large batches of documents need to be processed, posing a risk to the full realization of this new capability.

Recommendations

The digitization workflow can be enhanced and unified. The current workflow requires staff to manually move files between folders and double-click on scripts to execute, which can be inefficient and prone to error. Furthermore, the advanced LLM-based approach used for this bank's CTRs is more accurate than the older OCR methods used for other banks. Creating a single, user-friendly application for all digitization tasks would increase efficiency, reduce errors, and the highest quality data extraction for all historical documents.

III. AI-Powered Web Application for Compliance Report Review

Motivation

The FIU is mandated to review detailed compliance reports submitted by Financial Institutions (FIs) and Designated Non-Financial Businesses and Professions (DNFBPs). This process is traditionally manual, requiring officers to read through lengthy, dense documents to verify compliance against a standard checklist. This method is not only extremely time-consuming but is also susceptible to human error and inconsistency, potentially leading to missed compliance deficiencies.

There is a clear opportunity to leverage artificial intelligence to streamline this workflow. An AI-powered tool can automate the initial review process, rapidly locating relevant information and providing citations. This directly supports the FIU's mission by enabling more thorough, consistent, and efficient compliance monitoring, allowing staff to focus their expertise on analysis and enforcement rather than manual document review.

Outcomes

A custom web-based application was developed from scratch to serve as an AI assistant for compliance officers reviewing reports. The backend is built using the FastAPI framework, chosen for its easy integration with Python-based AI tools. The frontend is designed with standard HTML, CSS, and JavaScript, providing a clean and intuitive interface for non-technical users.

At the core of the system is a LangChain-based Retrieval-Augmented Generation (RAG) pipeline, which handles the logic for document parsing, embedding, retrieval, and LLM prompting. Uploaded documents (PDF, DOCX, or TXT) are parsed using existing Python libraries. For scanned documents, Tesseract OCR is integrated to extract text from image-based PDFs. Text chunks are then embedded and stored in a vector database.

When a user asks a compliance question, the app retrieves the most relevant passages and sends them to a locally hosted LLM via Ollama. The LLM is prompted to generate concise, cited answers.

Some key features were implemented, including:

- Upload of compliance documents in multiple formats (PDF, DOCX, TXT)
- Allows officers to create and manage reusable question lists, eliminating the need to retype
- An AI-powered question-answering function that automatically retrieves the most relevant excerpts from uploaded documents and generates an answer with citations referencing the supporting text
- OCR functionality was integrated to handle scanned (image-based) PDFs, expanding the tool's utility
- Docker is used for easier setup and deployment of the application

The FIU now has a powerful tool that significantly reduces the manual effort involved in compliance report reviews, resulting in faster and more consistent evaluations. During the project, the compliance officer Jeremy and the analyst Boboy actively tested and used the system. They uploaded multiple sample reports, ran some sessions, and verified that the responses aligned with the original report content.

Both users completed walkthroughs without resistance, demonstrating that the system can be used independently with minimal training. Their feedback helped fine-tune the system and improve the clarity of citation outputs. Source code and documentation have been provided to support continued use beyond the consulting period.

Similar to the digitization project, the primary limitation is performance due to server hardware. The AI features rely on the same CPU-only server, leading to slow response times when generating answers and citations.

Recommendations

The application currently displays citations as plain text. While functional, this requires the user to manually locate the cited text within the original document. Making the citations interactive, for example, by showing the source text in a pop-up window, would make the review process even faster and more seamless, directly improving workflow efficiency.

IV. Digital Forms for Compliance Information Submission

Motivation

The FIU's process for collecting annual compliance data from FIs and DNFBPs has historically relied on sending and receiving Excel spreadsheets via email. This method created significant operational inefficiencies, including inconsistent data formatting, manual data entry errors, and difficulty in aggregating information for risk analysis.

Transitioning to standardized digital forms provides a clear solution. This shift simplifies the submission process for reporting institutions while ensuring the FIU receives clean, structured, and immediately usable data, allowing for future integration with the risk analysis tool. This directly aligns with the FIU's mission to operate efficiently and maintain robust, up-to-date data to inform its national risk assessments.

Outcomes

A new digital data collection system was designed and implemented using the FIU's existing software suite.

After evaluating options, Zoho Forms was selected as the primary tool due to its integration with the FIU's existing Zoho Suite. I designed four distinct forms to capture policy and statistical data from both banks and DNFBPs, initially prototyping them in Google Forms before finalizing them in Zoho.

Two of the four designed forms were fully implemented and deployed in Zoho Forms, ready for use by the FIU. A video on how to manage the forms was provided to FIU staff.

This project modernized a critical data collection process, improving data quality and staff capacity. Data submitted via Zoho Forms is now standardized, eliminating formatting errors and inconsistencies. FIU staff now have the capacity to create, modify, and manage digital forms themselves, demonstrating a new organizational skill.

FIU-AML/CFT Banks - Data

1 Basic Information 2 A: Customer Base Information 3 B: Customer Risk Profile As Determined By The Bank Based On Internal Risk Assessment 4 C: Customer E According To Pl: Residence

Bank Name *

Reporting Date *

dd-MMM-yyyy

Contact Person *

First Name Last Name

Figure 1: Created Zoho Form

Currently, the FIU is currently using the free tier of Zoho Forms, which limits the organization to only three active forms and lacks key security features like field-level encryption. The form limit prevents the deployment of the other two designed forms. More importantly, the lack of security features on the free plan poses a potential risk when handling sensitive compliance information.

Recommendations

Upgrading to a paid Zoho Forms plan is a necessary investment to fully secure and scale the new data collection system. The Basic plan, at a minimal cost, removes the restrictive form limit and introduces essential security and privacy features. Once upgraded, the only required configuration is for the FIU to enable encryption on specific fields.

V. New Database Data Migration and Setup

Motivation

A new database for storing CTRs and STRs was installed at the Palau FIU in August 2024. However, shortly after the installation, the FIU became occupied with other priorities, such as relocating to new offices, which interrupted the continued use of the system. Subsequently, the FIU faced challenges importing data from August 2024 to July 2025 and configuring the system to operate within the new office's network environment. Without complete data migration and proper network setup, the FIU was unable to fully leverage the system's advanced analytical capabilities, limiting its effectiveness in supporting investigations.

Outcomes

To address the long-standing issues, I collaborated closely with Boboy to systematically identify and resolve the root causes that had prevented the FIU from using the new system since its installation in August 2024. First, we traced the system's inability to access its data import folders to incorrect IP address configurations that no longer matched the office's new network environment following their relocation. I reconfigured the network settings on the new system to align with the updated infrastructure, restoring file access and communication between systems.

Next, we focused on data migration. We initially did not know how to import data in bulk into the new system, as there was no documentation available. To resolve this, we arranged a meeting with the technician who had originally installed the system in August 2024 and learned the full import process from him. We successfully imported all the missing CTRs into the new system.

As a result, the new system is now fully operational and has begun to be integrated into the FIU's analytical workflow. Staff have successfully tested the upload process for new reports, and the system is now ready for routine use going forward.

However, the Microsoft Access database will continue to be used in parallel, as Boboy is more familiar with its interface and prefers it for certain tasks. For example, Access allows him to view all CTR records at once, whereas the new system displays only 20 records by default. This familiarity and functionality make Access more convenient for some basic operations.

Recommendations

Currently, staff must be physically present at the server machine to access the new database. This is inefficient and inconvenient. Configuring proper network access would allow authorized staff, like Boboy, to access the system directly from their own desktops. This would save time, improve workflow efficiency, and allow for easier system management.

VI. Additional Recommendations

Upgrade to GPU-enabled Server

This recommendation addresses a critical limitation identified across multiple projects. The FIU's current server is a CPU-only machine, which is inadequate for the intensive computational demands of the modern AI tools that have been implemented. I strongly recommended the FIU purchase and install a server equipped with a dedicated Graphics Processing Unit (GPU).

The success and long-term adoption of both the CTR digitization pipeline and the AI-powered compliance review application are currently at risk due to severe performance bottlenecks. For example, now the CTR digitization process takes about 20 minutes per document. A GPU-enabled server would reduce the time significantly. This makes batch processing feasible and encourages user adoption. Also, the AI compliance review application currently experiences lag. A GPU would provide near-instantaneous responses, making the tool a fluid and indispensable part of the daily workflow rather than a frustrating one.

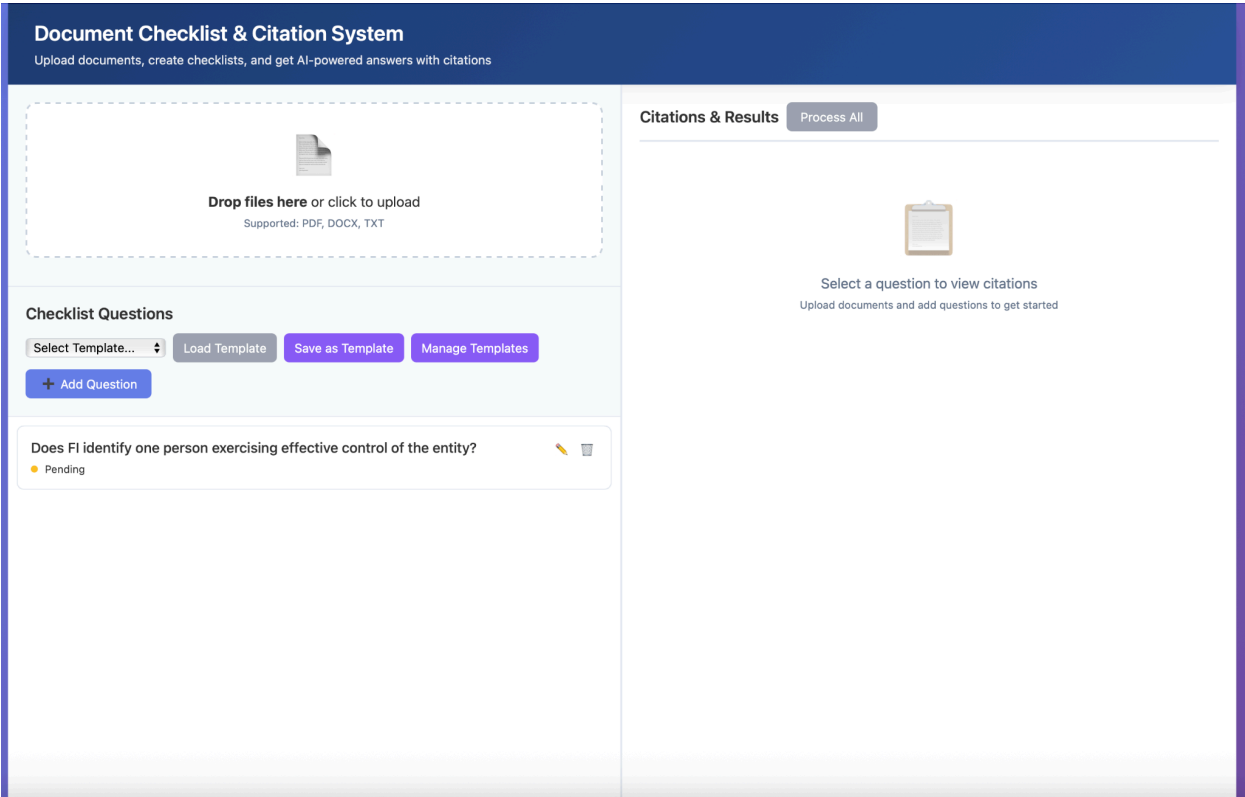
The FIU can consult with an IT vendor to select a server with a suitable GPU. At the moment, July 2025, a model like NVIDIA RTX 4080 or NVIDIA RTX 5070Ti offers an excellent balance of performance and cost for these types of AI workloads. The Python-based digitization pipeline and the FastAPI web application onto the new server, configuring them to utilize the GPU for all AI-related tasks.

About the Consultant

Yang Pan is a junior in Information Systems with an additional major in Computer Science at Carnegie Mellon University. He will be taking part in the Technology Consulting in the Global Community internship over the summer, and return in the fall to start senior year.

Appendix A.

The homepage of the AI-Powered Web Application for Compliance Report Review



Appendix B.

A screenshot of video instructions on how to edit Zoho Forms

